Ambrogio Maria Bernardelli

Curriculum Vitae et Studiorum

Research Positions

`01/2025` – `present` Postdoctoral Researcher, Department of Mathematics "F. Casorati", University of Pavia, Italy. Project: *Stochastic optimization for electrical energy storage systems and renewable energy sources.*

`09/2026` – `12/2026` Visiting Postdoctoral Research Associate, Department of Operations Research & Financial Engineering, Princeton University, Princeton, NJ 08544, USA. Visiting Prof. Bartolomeo Stellato's research group.

Education

Ph.D. in Computational Mathematics and Decision Sciences
University of Pavia and Università della Svizzera Italiana

Cycle: XXXVII.
Dates: 01/2022 – 12/2024.
Thesis title: *Methods for Combinatorial Optimization and Their Applications.*
Defended on: February 20, 2025.
Supervisor: Prof. Stefano Gualandi.

M.Sc. in Mathematics [LM-40]
University of Pavia

Dates: 10/2019 – 09/2021.
Thesis title: *Aspetti Algebrici e Combinatori della Teoria degli Operad* (Algebraic and Combinatorial Aspects of Operad Theory).
Defended on: September 23, 2021.
Supervisor: Prof. Alberto Canonaco.
Grade: 110/110 cum laude.

B.Sc. in Mathematics [L-35]
University of Pavia

Dates: 10/2016 – 09/2019.
Thesis title: *Rappresentazioni Lineari di Gruppi Finiti* (Linear Representations of Finite Groups).
Defended on: September 19, 2019.
Supervisor: Prof. Alberto Canonaco.
Grade: 110/110 cum laude.

Conferences Participation

Conferences and workshops where my works have been presented. Invited talks are labeled with †.

`06/2026` IPCO 2026, Padova, Italy. Poster.

`09/2025` The HEXAGON Workshop on power grids, Benevento, Italy†.

`09/2025` ODS 2025, Milan, Italy.

`07/2024` IPCO 2024, Wrocław, Poland. Poster.

07/2024 ISMP 2024 (presented by Simone Milanesi), Montréal, Canada†.

06/2024 EURO 2024 (presented by Simone Milanesi), Copenhagen, Denmark†.

06/2024 The HEXAGON Workshop on power grids, Bergamo, Italy†.

05/2024 ISCO 2024, San Cristóbal de La Laguna, Spain.

10/2023 2023 INFORMS Annual Meeting (presented by Stefano Gualandi), Phoenix, Arizona†.

06/2023 LION17, Nice, France. The work *The BeMi Stardust: a Structured Ensemble of Binarized Neural Network* presented here got accepted for the conference proceedings.

02/2023 7th AIROYoung Workshop (presented by Simone Milanesi), Milano, Italy.

02/2023 7th AIROYoung Workshop, Milano, Italy.

01/2023 The Mathematics of Machine Learning (presented by Simone Milanesi), Pisa, Italy.

11/2022 Matematica per l'Intelligenza Artificiale e il Machine Learning - Giovani ricercatori, Torino, Italy.

09/2022 YAMC 2022, Arenzano, Italy.

07/2022 ICCOPT / MOPTA 2022, Lehigh University, Bethlehem, Pennsylvania. The work *Scheduling elective surgeries under uncertainty: a multi-objective stochastic approach* presented here got the second place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

Publications

Journal Articles

[J3] Sosso, A., **Bernardelli, A. M.**, Gualandi, S. (2026). The Cloven Traveling Salesman: Cycle Covers and the Integrality Gap of Small ATSP Instances. Optimization Letters.

[J2] **Bernardelli, A. M.**, Gualandi, S., Milanesi, S., Lau, H. C., Yorke-Smith, N. (2024). Multiobjective Linear Ensembles for Robust and Sparse Training of Few-Bit Neural Network. INFORMS Journal on Computing.

[J1] **Bernardelli, A. M.**, Bonasera, L., Duma, D., Vercesi, E. (2024). Multi-objective stochastic scheduling of inpatient and outpatient surgeries. Flex Serv Manuf J.

Conference Proceedings

[P1] **Bernardelli, A. M.**, Gualandi, S., Lau, H.C., Milanesi, S. (2023). The BeMi Stardust: A Structured Ensemble of Binarized Neural Networks. In: Sellmann, M., Tierney, K. (eds) Learning and Intelligent Optimization. LION 2023. Lecture Notes in Computer Science, vol 14286. Springer, Cham.

Preprints

[A4] **Bernardelli, A. M.**, (2026). Cryptanalysis of operad-based cryptography. arXiv preprint arXiv:2611.10134. Currently under revision at Advances in Mathematics of Communications (AMC).

[A3] Sacilotto, A., **Bernardelli, A. M.**, Gualandi, S. (2026). Emission-aware operation of residential PV-battery systems: the role of information structure. arXiv preprint arXiv:2606.26648. Currently under revision at Sustainable Energy, Grids and Networks (SEGAN).

[A2] Riccardi, G., **Bernardelli, A. M.**, Gualandi, S. (2026). Theoretical Perspectives on Jabr-Type Convex Relaxations for AC Optimal Power Flow. arXiv preprint arXiv:2604.00664. Currently under revision at European Journal of Operational Research.

[A1] **Bernardelli, A. M.**, Vercesi, E., Gualandi, S., Mastrolilli, M., Gambardella, L. M. (2024). Lower bounds for the integrality gap of the bi-directed cut formulation of the Steiner Tree Problem. arXiv preprint arXiv:2405.13773. Currently under revision at Networks.

Awards

04/2023 Grant awarded by the TAILOR Connectivity Fund for a one-month visiting period at TU Delft, Delft, The Netherlands.

07/2022 2nd place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

09/2019 Department scholarship for the Master's degree program in Mathematics.

Visiting Periods

09/2023-02/2024 I spent six months in USI (Università della Svizzera italiana), Lugano, Switzerland, as part of my Ph.D. joint program, working on integrality gap problems with Professor Luca Maria Gambardella.

04/2023 I spent one month in TU Delft, Delft, The Netherlands, as a guest of STAR Lab, working on AI and optimization with Dr. Neil Yorke-Smith.

Conference Organization

05/2025 I was one of the organizers of the COMPMAT Spring Workshop 2025, that was held at University of Pavia on May 23, 2025.

02/2025 I was part of the organizing committee of the 9th AIROYoung Workshop, that was held at University of Pavia from the 26th to the 28th of February, 2025.

2022-2024 I was a co-organizer of Caffè Beltrami, a cycle of seminars aimed at introducing different areas of research in mathematics to Bachelor's and Master's students.

Teaching Activities

Teaching

2026/2027 511180 – Algorithms for Optimization (M.Sc. in Finance, LM-16), University of Pavia, 22 hours. Co-teaching with Prof. Davide Duma, who held the other 22 hours of the course.

2024/2025 511180 – Algorithms for Optimization (M.Sc. in Finance, LM-16), University of Pavia, 22 hours. Co-teaching with Prof. Daniele Boffi, who held the other 22 hours of the course.

Supervising

2025/2026 *Ottimizzazione Stocastica Multistadio per la Gestione del Rischio nel Mercato Energetico Italiano*, Silvia Maria Teresa Stimolo, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2024/2025 *Riduzione delle Emissioni di Carbonio mediante Ottimizzazione Stocastica dello Stoccaggio Energetico a Batteria in Sistemi Fotovoltaici*, Anna Sacilotto, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2024/2025 *Strengthening the SOCP formulation for the ACOPF problem*, Arthur Mazeyrat, Master's Report in Operations Research and Combinatorial Optimization at University Grenoble Alpes, co-supervised with Prof. Stefano Gualandi and Gabor Riccardi.

2023/2024 *The Cloven Travelling Salesman: a new Approach to the ATSP Integrality Gap Estimation*, Alessandro Sosso, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *On the exactness of Jabr-like models and distributionally robust stochastic optimal power flow*, Gabor Riccardi, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *Analisi di architetture per Binarized Neural Networks*, Andrea Panno, Bachelor's Thesis in Bioengineering at University of Pavia, co-supervised with Prof. Stefano Gualandi and Simone Milanesi.

Tutoring

2021/2022 Tutoring activity for the course *Geometria e Algebra* that was held by prof. Francesco Bonsante for the Bachelor's degree program *Bioingegneria* at University of Pavia. 40 hours.

2021/2022 Tutoring activity for first-year students attending the Bachelor's degree program in Mathematics at the University of Pavia. 40 hours.

Personal Information

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